

Below is a ****complete Oracle Index example script**** with tables, sample data, indexes, and testing queries.

```
``sql
/* =====
ORACLE INDEX COMPLETE PRACTICE SCRIPT
Scenario: Telecom Subscriber & Call Detail Records
===== */

/* =====
1. CLEAN OLD OBJECTS
===== */

BEGIN
  EXECUTE IMMEDIATE 'DROP TABLE call_records CASCADE CONSTRAINTS';
EXCEPTION
  WHEN OTHERS THEN NULL;
END;
/

BEGIN
  EXECUTE IMMEDIATE 'DROP TABLE subscribers CASCADE CONSTRAINTS';
EXCEPTION
  WHEN OTHERS THEN NULL;
END;
/

/* =====
2. CREATE TABLES
===== */

CREATE TABLE subscribers
(
  subscriber_id  NUMBER PRIMARY KEY,
  msisdn        VARCHAR2(20) NOT NULL,
  customer_name  VARCHAR2(100),
  nrc_no        VARCHAR2(50),
  city          VARCHAR2(50),
```

```
status      VARCHAR2(20),
activation_date DATE
);
```

```
CREATE TABLE call_records
(
  call_id      NUMBER PRIMARY KEY,
  subscriber_id NUMBER,
  called_number VARCHAR2(20),
  call_type    VARCHAR2(20),
  call_start   DATE,
  duration_sec NUMBER,
  charge_amount NUMBER(10,2),
  CONSTRAINT fk_call_subscriber
  FOREIGN KEY (subscriber_id)
  REFERENCES subscribers(subscriber_id)
);
```

```
/* =====
3. INSERT SAMPLE DATA
===== */
```

```
INSERT INTO subscribers VALUES
(1, '959400000001', 'Aung Aung', '12/ABC(N)123456', 'Yangon', 'ACTIVE', DATE '2023-01-10');
```

```
INSERT INTO subscribers VALUES
(2, '959400000002', 'Mg Mg', '12/DEF(N)234567', 'Mandalay', 'ACTIVE', DATE '2023-02-15');
```

```
INSERT INTO subscribers VALUES
(3, '959400000003', 'Hla Hla', '12/GHI(N)345678', 'Naypyitaw', 'INACTIVE', DATE '2023-03-20');
```

```
INSERT INTO subscribers VALUES
(4, '959400000004', 'Su Su', '12/JKL(N)456789', 'Yangon', 'ACTIVE', DATE '2023-04-25');
```

```
INSERT INTO subscribers VALUES
(5, '959400000005', 'Kyaw Kyaw', '12/MNO(N)567890', 'Bago', 'SUSPENDED', DATE '2023-05-30');
```

```
INSERT INTO call_records VALUES
(101, 1, '959500000001', 'VOICE', DATE '2024-01-01', 120, 150.00);
```

```
INSERT INTO call_records VALUES
(102, 1, '959500000002', 'VOICE', DATE '2024-01-02', 300, 350.00);
```

```
INSERT INTO call_records VALUES
(103, 2, '959500000003', 'SMS', DATE '2024-01-03', 0, 15.00);
```

```
INSERT INTO call_records VALUES
(104, 3, '959500000004', 'DATA', DATE '2024-01-04', 600, 500.00);
```

```
INSERT INTO call_records VALUES
(105, 4, '959500000005', 'VOICE', DATE '2024-01-05', 180, 200.00);
```

```
INSERT INTO call_records VALUES
(106, 4, '959500000006', 'SMS', DATE '2024-01-06', 0, 15.00);
```

```
INSERT INTO call_records VALUES
(107, 5, '959500000007', 'DATA', DATE '2024-01-07', 900, 800.00);
```

```
COMMIT;
```

```
/* =====
4. BASIC TEST BEFORE INDEX
===== */
```

```
SQL> SELECT *
FROM subscribers
WHERE msisdn = '9594000000004';
      2      3
```

```
SUBSCRIBER_ID MSISDN
```

```
-----
```

```
CUSTOMER_NAME
```

```
-----
```

```
NRC_NO
```

```
-----
```

```
CITY
```

```
STATUS
```

```
-----
```

```
ACTIVATIO
```

```
-----
```

```
4 9594000000004
```

```
Su Su
```

```
12/JKL(N)456789
```

```
SUBSCRIBER_ID MSISDN
```

```
-----
```

```
CUSTOMER_NAME
```

```
-----
```

```
NRC_NO
```

```
-----
```

```
CITY
```

```
STATUS
```

```
-----
```

```
ACTIVATIO
```

```
-----
```

```
Yangon
```

```
ACTIVE
```

```
25-APR-23
```

```
SQL> SELECT *
FROM subscribers
WHERE city = 'Yangon';
```

2	3		
SUBSCRIBER_ID	MSISDN		
-----	-----		
CUSTOMER_NAME			

NRC_NO			

CITY		STATUS	
-----		-----	
ACTIVATIO			

	1 9594000000001		
Aung Aung			
12/ABC(N)123456			
SUBSCRIBER_ID	MSISDN		
-----	-----		
CUSTOMER_NAME			

NRC_NO			

CITY		STATUS	
-----		-----	
ACTIVATIO			

Yangon		ACTIVE	
10-JAN-23			

```
SQL> SELECT *
FROM call_records
WHERE subscriber_id = 4;
```

2	3			
CALL_ID	SUBSCRIBER_ID	CALLED_NUMBER	CALL_TYPE	CALL_STAR
-----	-----	-----	-----	-----
DURATION_SEC	CHARGE_AMOUNT			
-----	-----			
105	4	9595000000005	VOICE	05-JAN-24
180	200			
106	4	9595000000006	SMS	06-JAN-24
0	15			

```
SQL> SELECT *
FROM call_records
WHERE call_start BETWEEN DATE '2024-01-01' AND DATE '2024-01-05';
```

CALL_ID	SUBSCRIBER_ID	CALLED_NUMBER	CALL_TYPE	CALL_STAR
101	1	959500000001	VOICE	01-JAN-24
120	150			
102	1	959500000002	VOICE	02-JAN-24
300	350			
103	2	959500000003	SMS	03-JAN-24
0	15			
104	3	959500000004	DATA	04-JAN-24
600	500			
105	4	959500000005	VOICE	05-JAN-24
180	200			

```
/* =====
5. CREATE INDEXES
===== */
```

```
/* Unique index for phone number lookup */
CREATE UNIQUE INDEX idx_subscribers_msisdn
ON subscribers(msisdn);
```

```
/* Normal B-tree index for city search */
CREATE INDEX idx_subscribers_city
ON subscribers(city);
```

```
/* Composite index for city and status search */
CREATE INDEX idx_subscribers_city_status
ON subscribers(city, status);
```

```
/* Foreign key index for faster join */
CREATE INDEX idx_call_records_subscriber_id
ON call_records(subscriber_id);
```

```
/* Date index for call date filtering */  
CREATE INDEX idx_call_records_call_start  
ON call_records(call_start);
```

```
/* Composite index for reporting */  
CREATE INDEX idx_call_records_type_date  
ON call_records(call_type, call_start);
```

```
/* Function-based index for case-insensitive search */  
CREATE INDEX idx_subscribers_upper_name  
ON subscribers(UPPER(customer_name));
```

```

/* =====
6. TEST AFTER INDEX
===== */

```

```

/* Test 1: Unique index usage */

```

```

SQL> SELECT *
FROM subscribers
WHERE msisdn = '959400000004';
      2      3
SUBSCRIBER_ID MSISDN
-----
CUSTOMER_NAME
-----
NRC_NO
-----
CITY STATUS
-----
ACTIVATIO
-----
      4 959400000004
Su Su
12/JKL(N)456789

SUBSCRIBER_ID MSISDN
-----
CUSTOMER_NAME
-----
NRC_NO
-----
CITY STATUS
-----
ACTIVATIO
-----
Yangon ACTIVE
25-APR-23

```

/* Test 2: City index usage */

```
SQL> SELECT *  
FROM subscribers  
WHERE city = 'Yangon';
```

```
2      3  
SUBSCRIBER_ID MSISDN
```

CUSTOMER_NAME

NRC_NO

CITY STATUS

ACTIVATIO

1 9594000000001

Aung Aung

12/ABC(N)123456

SUBSCRIBER_ID MSISDN

CUSTOMER_NAME

NRC_NO

CITY STATUS

ACTIVATIO

Yangon

ACTIVE

10-JAN-23

/* Test 3: Composite index usage */

```
SQL> SELECT *  
FROM subscribers  
WHERE city = 'Yangon'  
AND status = 'ACTIVE';  
      2      3      4  
SUBSCRIBER_ID MSISDN
```

CUSTOMER_NAME

NRC_NO

CITY STATUS

ACTIVATIO

1 9594000000001

Aung Aung
12/ABC(N)123456

SUBSCRIBER_ID MSISDN

CUSTOMER_NAME

NRC_NO

CITY STATUS

ACTIVATIO

Yangon ACTIVE
10-JAN-23

/* Test 4: Foreign key index usage */

```
SQL> SELECT s.customer_name,
           s.msisdn,
           c.call_type,
           c.duration_sec,
           c.charge_amount
FROM subscribers s
JOIN call_records c
ON s.subscriber_id = c.subscriber_id
WHERE c.subscriber_id = 4;
```

CUSTOMER_NAME	MSISDN	CALL_TYPE	DURATION_SEC	CHARGE_AMOUNT
Su Su	959400000004	VOICE	180	200
Su Su	959400000004	SMS	0	15

/* Test 5: Date index usage */

```
SQL> SELECT *
FROM call_records
WHERE call_start BETWEEN DATE '2024-01-01' AND DATE '2024-01-05';
```

CALL_ID	SUBSCRIBER_ID	CALLED_NUMBER	CALL_TYPE	CALL_STAR
101	1	959500000001	VOICE	01-JAN-24
120	150			
102	1	959500000002	VOICE	02-JAN-24
300	350			
103	2	959500000003	SMS	03-JAN-24
0	15			
104	3	959500000004	DATA	04-JAN-24
600	500			
105	4	959500000005	VOICE	05-JAN-24
180	200			

/* Test 6: Composite reporting index */

```
SQL> SELECT *
FROM call_records
WHERE call_type = 'VOICE'
AND call_start >= DATE '2024-01-01';
```

2	3	4			
CALL_ID	SUBSCRIBER_ID	CALLED_NUMBER	CALL_TYPE	CALL_STAR	
-----	-----	-----	-----	-----	
DURATION_SEC	CHARGE_AMOUNT				
-----	-----				
101	1	9595000000001	VOICE	01-JAN-24	
120	150				
102	1	9595000000002	VOICE	02-JAN-24	
300	350				
105	4	9595000000005	VOICE	05-JAN-24	
180	200				

```
/* Test 7: Function-based index */
```

```
SQL> SELECT *
FROM subscribers
WHERE UPPER(customer_name) = 'SU SU';
```

```
2      3
SUBSCRIBER_ID MSISDN
```

```
-----
```

```
CUSTOMER_NAME
```

```
-----
```

```
NRC_NO
```

```
-----
```

```
CITY
```

```
STATUS
```

```
-----
```

```
ACTIVATIO
```

```
-----
```

```
4 9594000000004
```

```
Su Su
```

```
12/JKL(N)456789
```

```
SUBSCRIBER_ID MSISDN
```

```
-----
```

```
CUSTOMER_NAME
```

```
-----
```

```
NRC_NO
```

```
-----
```

```
CITY
```

```
STATUS
```

```
-----
```

```
ACTIVATIO
```

```
-----
```

```
Yangon
```

```
ACTIVE
```

```
25-APR-23
```

```
/* =====
```

```
7. CHECK EXECUTION PLAN
```

```
===== */
```

```
SQL> EXPLAIN PLAN FOR
SELECT *
FROM subscribers
WHERE msisdn = '9594000000004';
2      3      4
Explained.
```

```

SQL> SELECT *
FROM TABLE(DBMS_XPLAN.DISPLAY);
2
PLAN_TABLE_OUTPUT
-----
Plan hash value: 2169606609

-----

| Id | Operation | Name | Rows | Bytes | Cost (%CPU)| Time |
-----
PLAN_TABLE_OUTPUT
-----
| 0 | SELECT STATEMENT | | 1 | 152 | |
1 (0)| 00:00:01 |
| 1 | TABLE ACCESS BY INDEX ROWID | SUBSCRIBERS | 1 | 152 | |
1 (0)| 00:00:01 |
|* 2 | INDEX UNIQUE SCAN | IDX_SUBSCRIBERS_MSISDN | 1 | | |
0 (0)| 00:00:01 |
-----
PLAN_TABLE_OUTPUT
-----

Predicate Information (identified by operation id):
-----

2 - access("MSISDN"='9594000000004')

14 rows selected.

```

```

SQL> EXPLAIN PLAN FOR
SELECT *
FROM subscribers
WHERE city = 'Yangon'
AND status = 'ACTIVE';
2 3 4 5
Explained.

```

```

SQL> SELECT *
FROM TABLE(DBMS_XPLAN.DISPLAY);
2
PLAN_TABLE_OUTPUT
-----
Plan hash value: 1332594611

-----
-
| Id | Operation          | Name          | Rows | Bytes | Cost (%CPU)| Time |
|-----|-----|-----|-----|-----|-----|-----|
|
-----
-

PLAN_TABLE_OUTPUT
-----
| 0 | SELECT STATEMENT |              | 2 | 304 | 2 (0)| 00:00:01 |
|
|* 1 | TABLE ACCESS FULL| SUBSCRIBERS | 2 | 304 | 2 (0)| 00:00:01 |
|
-----
-

Predicate Information (identified by operation id):

PLAN_TABLE_OUTPUT
-----
-----
1 - filter("CITY"='Yangon' AND "STATUS"='ACTIVE')

Note
-----
- dynamic statistics used: dynamic sampling (level=2)

17 rows selected.

```

```

SQL> EXPLAIN PLAN FOR
SELECT *
FROM call_records
WHERE call_type = 'VOICE'
AND call_start >= DATE '2024-01-01';
2      3      4      5
Explained.

```

```
SQL> SELECT *
FROM TABLE(DBMS_XPLAN.DISPLAY);
2
PLAN_TABLE_OUTPUT
```

```
-----
Plan hash value: 3625081644
```

```
--
```

Id	Operation	Name	Rows	Bytes	Cost (%CPU)	Time
----	-----------	------	------	-------	-------------	------

```
--
```

```
PLAN_TABLE_OUTPUT
```

0	SELECT STATEMENT		3	255	2 (0)	00:00:01
* 1	TABLE ACCESS FULL	CALL_RECORDS	3	255	2 (0)	00:00:01

```
--
```

```
Predicate Information (identified by operation id):
```

```
PLAN_TABLE_OUTPUT
```

```
-----
1 - filter("CALL_TYPE"='VOICE' AND "CALL_START">=TO_DATE(' 2024-01-01
00:00:00', 'syyyymm-dd hh24:mi:ss'))
```

```
Note
```

```
-----
```

```
- dynamic statistics used: dynamic sampling (level=2)
```

```
18 rows selected.
```

```

/* =====
8. VIEW ALL INDEXES
===== */

```

```

SQL> SELECT index_name,
           table_name,
           uniqueness,
           status
FROM user_indexes
WHERE table_name IN ('SUBSCRIBERS', 'CALL_RECORDS')
ORDER BY table_name, index_name;
  2      3      4      5      6      7

```

```
INDEX_NAME
```

```
-----
TABLE_NAME
```

```
-----
UNIQUENES STATUS
```

```
-----
IDX_CALL_RECORDS_CALL_START
CALL_RECORDS
NONUNIQUE VALID

```

```

IDX_CALL_RECORDS_SUBSCRIBER_ID
CALL_RECORDS
NONUNIQUE VALID

```

```
INDEX_NAME
```

```
-----
TABLE_NAME
```

```
-----
UNIQUENES STATUS
```

```
-----
IDX_CALL_RECORDS_TYPE_DATE
CALL_RECORDS
NONUNIQUE VALID

```

```

SYS_C008362
CALL_RECORDS

```

```

/* =====
9. VIEW INDEX COLUMNS
===== */

```

```

SQL> SELECT index_name,
           table_name,
           column_name,
           column_position
FROM user_ind_columns
WHERE table_name IN ('SUBSCRIBERS', 'CALL_RECORDS')
ORDER BY index_name, column_position;

```

```

2      3      4      5      6      7
INDEX_NAME
-----
TABLE_NAME
-----
COLUMN_NAME
-----
COLUMN_POSITION
-----
IDX_CALL_RECORDS_CALL_START
CALL_RECORDS
CALL_START
1

```

```

INDEX_NAME
-----
TABLE_NAME
-----
COLUMN_NAME
-----
COLUMN_POSITION
-----
IDX_CALL_RECORDS_SUBSCRIBER_ID
CALL_RECORDS
SUBSCRIBER_ID
1

```

```

/* =====
10. INDEX MONITORING
===== */

```

```

ALTER INDEX IDX_SUBSCRIBERS_MSISDN MONITORING USAGE;

```

```
SQL> SELECT *
FROM USER_OBJECT_USAGE
WHERE INDEX_NAME = 'IDX_SUBSCRIBERS_MSISDN';
  2      3
INDEX_NAME
-----
TABLE_NAME
-----
MON USE START_MONITORING      END_MONITORING
--  --  -----
IDX_SUBSCRIBERS_MSISDN
SUBSCRIBERS
YES NO  06/09/2026 05:36:32
```

```
SQL> SELECT * FROM V$VERSION;

BANNER
-----
BANNER_FULL
-----
BANNER_LEGACY
-----
      CON_ID
      -----
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
      0
```

```
SELECT *
FROM USER_INDEX_USAGE;
```

```
SELECT *
FROM DBA_OBJECT_USAGE
WHERE INDEX_NAME = 'IDX_SUBSCRIBERS_MSISDN';
```

== should be DBA role ==

```
ALTER INDEX idx_subscribers_msisdn MONITORING USAGE;
```

```
SQL> SELECT *
FROM subscribers
WHERE msisdn = '959400000004';
```

```
2      3
SUBSCRIBER_ID MSISDN
```

```
-----
```

```
CUSTOMER_NAME
```

```
-----
```

```
NRC_NO
```

```
-----
```

```
CITY
```

```
STATUS
```

```
-----
```

```
ACTIVATIO
```

```
-----
```

```
4 959400000004
```

```
Su Su
```

```
12/JKL(N)456789
```

```
SUBSCRIBER_ID MSISDN
```

```
-----
```

```
CUSTOMER_NAME
```

```
-----
```

```
NRC_NO
```

```
-----
```

```
CITY
```

```
STATUS
```

```
-----
```

```
ACTIVATIO
```

```
-----
```

```
Yangon
```

```
ACTIVE
```

```
25-APR-23
```

```
SELECT *
FROM v$object_usage
WHERE index_name = 'IDX_SUBSCRIBERS_MSISDN';
```

=== should be DBA role ==

```
/* Stop monitoring */
```

```
ALTER INDEX idx_subscribers_msisdn NOMONITORING USAGE;
```

```
/* =====  
11. REBUILD INDEX  
===== */
```

```
ALTER INDEX idx_subscribers_msisdn REBUILD;
```

```
/* =====  
12. DROP INDEX EXAMPLE  
===== */
```

```
-- DROP INDEX idx_subscribers_city;
```

```
/* =====  
13. IMPORTANT NOTES  
=====
```

1. Primary key automatically creates an index.
2. Unique index prevents duplicate values.
3. Foreign key columns should usually be indexed.
4. Indexes improve SELECT performance.
5. Too many indexes slow down INSERT, UPDATE, DELETE.
6. Composite index column order is important.
7. Function-based index works only when query uses same function.
8. Low-cardinality columns like STATUS may not always benefit from normal index.
9. Always check execution plan.
10. Indexes are best for large tables, not very small tables.

```
===== */  
...
```

This script is ready to run in **Oracle SQL Developer / SQL*Plus**.